The Low-Income Household Cost Burden Dashboard

Energy, Food, and Labor Equity in the United States

Vincent Alulu | PID A59024801 | Global Macroeconomic Data Analysis | Spring 2026
Prof. Kyle Handley | UC San Diego

Dashboard Link: https://vhalulu-2020.shinyapps.io/household-cost-burden/

1. Question and Policy Motivation

This project addresses an important question in economic policy: are low-income U.S. households losing purchasing power as food and energy prices rise faster than wages? Although energy inflation, food price increases, and wage stagnation have each been widely studied, their combined effects on low-income households are rarely examined in one accessible, real-time tool. This dashboard fills that gap.

The policy motivation is clear. The federal minimum wage has remained at $7.25 per hour since 2009, making it the longest period without an increase in U.S. history. As inflation has risen, the real value of that wage has fallen substantially. At the same time, low-income households spend a much larger share of their income (6-8%) on energy than higher-income households (1-2%), (ACEEE, 2020; U.S. Energy Information Administration, 2023), making them especially vulnerable to rising utility and fuel costs. Food prices have also increased sharply since 2020, with restaurant and fast-food prices rising even faster than grocery prices.

These challenges are often studied separately. The contribution of this dashboard is to bring them together into a single interactive, policy-facing tool that allows users to monitor the combined affordability challenge on low-income households in real time.

The primary intended user is a hypothetical policy analyst at the Center for Equitable Energy and Economic Policy (CEEEP), a U.S.-based NGO advising legislators on economic conditions facing low-income households, though the dashboard is designed to be accessible to journalists, researchers, and students without requiring deep econometric knowledge.

2. Data Sources and Key Indicators

All data are obtained through the FRED API maintained by the Federal Reserve Bank of St. Louis, (FRED Economic Data). The data are publicly available and may be freely used with attribution. The dashboard covers the period of January 2000 through April 2026, providing up to 26 years of monthly observations. Most series are reported monthly. Weekly gasoline prices are converted to monthly averages using the arithmetic means of all weekly observations within each month.

The variables are grouped into three categories:

Energy: The Consumer Price Index for All Energy (CPIENGSL) measures overall household energy inflation, including gasoline, electricity, natural gas, and heating oil. Regular gasoline prices (GASREGCOVW) are included as a supplementary indicator and converted from weekly to monthly averages.

Food: The CPI for Food at Home (CPIUFDSL) measures grocery price inflation, including items such as bread, meat, dairy products, fruits, and packaged foods. The CPI for Food Away from Home (CUUR0000SEFV) tracks prices at restaurants and fast-food outlets. To capture upstream cost pressures, the dashboard also includes the Producer Price Index for Farm Products (WPU01) and the Producer Price Index for Grains (WPU0111), which often signal future changes in consumer food prices.

Labor: Average hourly earnings in the Leisure and Hospitality sector (CES7000000003) are used as a proxy for low-income wage growth because it is one of the lowest-paying major sectors in the U.S.

economy. Average hourly earnings for all private employees (CES0500000003) provide a benchmark for comparison. Unemployment rates for workers with less than a high school education (LNS14027659) and those with a bachelor's degree or higher (LNS14027689) are included to examine labor market inequality. The federal minimum wage (FEDMINNFRWG) and headline CPI (CPIAUCSL) are used to calculate real minimum wages. NBER recession dates (USREC) are used to identify recession periods in all visualizations.

Five **transformations** are applied to the raw data before visualization. First, the real minimum wage is calculated by adjusting the federal minimum wage for inflation using the CPI. The series is rebased so that 2009, the last year the federal minimum wage was increased, equals 100. This allows changes in purchasing power to be tracked over time. Second, a 12-month trailing moving average is applied to all series using a rolling mean. This smooths short-term fluctuations and highlights longer-term trends. Third, the Education Unemployment Gap is calculated as the difference between the unemployment rate of workers without a high school diploma and that of workers with a bachelor's degree or higher. This provides a simple measure of labor market inequality. Fourth, the Household Affordability Gap Index, the dashboard's main original measure, is calculated as the average of food-at-home inflation and energy inflation minus wage growth in the Leisure and Hospitality sector. All components are expressed as 12-month percentage changes. Positive values indicate that living costs are rising faster than wages (shaded in red), while negative values indicate that wages are growing faster than costs (shaded in green). Fifth, a Low-Income Sahm Indicator is created by adapting the Sahm (2019) recession rule for workers without a high school diploma. It is calculated as the difference between the three-month average unemployment rate and its lowest value over the previous 12 months. A value of 0.5 percentage points or higher then national rate signals a potential labor market downturn.

3. Dashboard Design and User Interaction

The dashboard is built in R Shiny using the **bslib** package for modern layout and bootstrap styling. It is publicly deployed on ShinyApps.io and organized into five interactive tabs, each focusing on a different aspect of the cost burden facing low-income households.

Home: The dashboard opens with an About section that frames the policy context for all users: 'This dashboard explores the growing financial pressure facing low-income households in the United States, focusing on how rising food and energy prices, combined with wages that have not kept pace with inflation, are reducing purchasing power and making everyday essentials less affordable. It then displays four headline indicators: the unemployment rate, CPI inflation, energy inflation, and the real minimum wage in 2009 dollars. A three-panel "Why This Matters" section provides policy context for non-specialist users. **Energy Burden:** Shows energy inflation or gasoline prices alongside Leisure and Hospitality wage growth. Red shading highlights periods when energy costs rise faster than wages, while green shading indicates periods when wages outpace energy costs.

Food Economics: Allows users to explore food-at-home inflation, food-away-from-home inflation, farm product prices, and grain prices. Recession periods and moving averages can be displayed for additional context. **Labor Markets:** Includes unemployment rates by education level, the Education Unemployment Gap, real and nominal wage growth, the real minimum wage, and the Low-Income Sahm Indicator. **The Cost Burden:** Displays the Household Affordability Gap Index, the dashboard's main original measure. Red bars indicate periods when living costs rise faster than wages, while green bars indicate periods when wages keep pace with or exceed costs. Each tab includes controls for selecting a date range, applying a 12-month moving average, switching to a log scale, and downloading the data as a CSV file with a FRED citation header. A collapsible methodology section beneath each chart explains the data and calculations used.

To keep the dashboard current, the application automatically checks the age of its stored data each time it launches. If the data are more than 30 days old, all series are automatically refreshed from FRED.

4. Main Insights and Policy Takeaways

The real minimum wage has fallen sharply since 2009. Although the federal minimum wage remains $7.25 per hour, its purchasing power has declined substantially. By April 2026, it is worth about $4.68 in 2009 dollars, a decline of roughly 35 percent. The dashboard clearly illustrates how a fixed nominal wage loses value over time as prices rise, highlighting the case for indexing the minimum wage to inflation.

Energy price shocks place a disproportionate burden on low-income households. The Energy Burden tab shows that during major energy shocks, including 2008, 2020, and 2022, energy prices increased much faster than wages in the Leisure and Hospitality sector. These periods appear clearly in the dashboard and demonstrate how rising energy costs can quickly reduce household purchasing power. The 2022 surge was visible well before its peak, suggesting that real-time monitoring could support earlier policy responses.

Low-income workers experience more severe labor market downturns. The Low-Income Sahm Indicator shows much larger increases during recessions than the standard national measure. The indicator rose by about 2 percentage points during the 2001 recession, 6 percentage points during the 2008-09 financial crisis, and 14 percentage points during the COVID-19 shock. These results suggest that national indicators often understate the severity of downturns for the most vulnerable workers.

The Household Affordability Gap Index provides a useful real-time measure of financial pressure. The index was strongly positive during the 2008 energy shock and again in 2021-2022 when inflation outpaced wage growth. It remained close to zero during much of the 2010-2019 recovery period. By April 2026, the index had returned to near zero, indicating some relief from recent inflation pressures. However, this does not fully offset the cumulative loss in purchasing power since 2009. A practical policy recommendation is to monitor the index regularly and consider targeted food and energy assistance when it rises above five percentage points.

5. Limitations and Future Improvements

The following limitations should be noted.

Proxy variables rather than direct income data. Because FRED data are mostly aggregate, the dashboard uses Leisure and Hospitality wages and education-level unemployment as proxies for low-income conditions. These are standard in applied policy research but do not directly measure low-income household income or spending. A stronger approach would use microdata from the Bureau of Labor Statistics Consumer Expenditure Survey (Consumer Expenditure Survey), which reports spending patterns by income quintile. The CEX API was considered but not included due to implementation complexity.

Energy burden by income group not included. The original design included an energy expenditure share by income quintile, showing that the bottom quintile spends about 6 to 8 percent of income on energy compared to 1 to 2 percent for the top quintile. This was not implemented due to data and API constraints, but it would significantly strengthen the analysis.

No causal identification. The dashboard is descriptive. It shows how food and energy costs move relative to wages, but it does not estimate causal effects or counterfactual outcomes. A formal econometric model would be required for causal inference.

National-level data only. All series are national aggregates. This masks important variation across states and cities, where energy prices, food costs, and wages differ substantially. Future versions could incorporate state-level FRED data to capture geographic inequality.

Update frequency limitations. The dashboard refreshes data on launch and re-pulls from FRED if more than 30 days have passed. As a result, it may lag up to one month behind the most recent available data.

Despite these limitations, the dashboard provides a useful policy tool. It combines long-run macroeconomic data across three domains, applies five original transformations, and presents them in an accessible interactive format. The Household Affordability Gap Index and the Low-Income SAHM Indicator are particularly novel and extend techniques learnt from the course, including moving averages and labor market indicators into a practical setting.

References

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